

Yabetse Tesfaye Alemu

Computer Science Student

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 Portfolio  Leetcode

EDUCATION

Bachelor of Computer Science

American College of Technology 

03/2023 – Present
Addis Ababa, Ethiopia

Relevant Coursework: Advanced Database, Data Structures & Algorithms, Web & Mobile Development, Machine Learning & AI, Distributed Systems, Operating Systems & System Programming

Key Skills & Experience:

- Built full-stack applications with responsive front-end web design (HTML5, CSS3, JavaScript, React) and modern backend development (Node.js, Python, REST APIs)
- Applied DevOps practices including version control (Git), CI/CD pipelines, containerization (Docker), and basic cloud deployment (AWS/GCP)
- Developed cross-platform mobile applications with a focus on performance, UI/UX, and offline data sync
- Designed and implemented distributed system components, multithreading, and inter-process communication in system programming projects
- Leveraged machine learning models and data processing pipelines for AI-driven features in academic and personal projects

PROFESSIONAL EXPERIENCE

LLM Model Trainer and Data Annotator

Turing

07/2025 – 11/2025
Remote

- Trained LLMs by manually annotating video data and writing precise, time-frame-level descriptions to help the model understand context and actions per second
- Shifted from model trainer to data annotator to improve training data quality, ensuring the AI learns accurate temporal patterns
- Tested model responses by designing prompts and verifying output correctness against expected behavior, closing the feedback loop for continuous improvement

Frontend Developer

Upwork

03/2025 – 05/2025
Remote

- Developed a high-performance frontend for the Chessline website using Next.js, ensuring fast page loads and smooth navigation
- Translated the company’s design vision from Figma into a fully responsive, pixel-perfect layout
- Implemented SEO best practices (meta tags, structured data, performance optimization) to improve search visibility
- Fixed cross-device responsiveness issues, delivering a consistent user experience across desktop, tablet, and mobile

AI Infrastructure & DevOps Training Engineer

Revelo

03/2026 – 06/2026
Remote

- Worked at Revelo on Infrastructure as Code (IaC) and AI training tasks focused on improving the reasoning and code generation capabilities of Amazon Nova Lite v2 models.
- Created and evaluated complex technical tasks involving YAML, JSON, Python, AWS CDK (Java), Terraform, and cloud infrastructure workflows.
- Designed high-difficulty prompts, validation scenarios, and infrastructure configurations to train and benchmark AI systems on real-world DevOps and cloud engineering problems.
- Contributed to improving model accuracy in infrastructure automation, configuration management, and cloud deployment reasoning.

PROJECTS

ImmuStore DB

Distributed Database Engine Built from Scratch

- Engineered a distributed database engine from scratch in pure Python, featuring a copy-on-write B+ tree storage core with $O(\log n)$ reads, writes, sorted scans, lazy record loading, and append-only persistence with no third-party runtime dependencies.
- Implemented crash-safe durability and MVCC using shadow-paging commits, CRC32 record validation, double-buffered metadata blocks, torn-write recovery, lock-free point-in-time snapshots, and optimistic transactions with write-conflict detection.
- Built a Redis-compatible asynchronous network server using RESP and asyncio, supporting standard Redis clients, real-time prefix-based change data capture, concurrent connections, and live set/delete event streaming.
- Developed distributed replication through a from-scratch Raft consensus implementation with leader election, replicated logs, automatic failover, TCP cluster communication, and exactly-once application of committed operations.
- Added document collections, secondary indexes, query planning, TTL expiration, Prometheus metrics, administrative health endpoints, Docker deployment, and deterministic fault-injection testing across crashes, network partitions, corrupted metadata, and torn disk writes.

Tech stacks: Python, asyncio, B+ Trees, MVCC, Raft Consensus, Redis RESP, TCP Networking, SQLite, Prometheus, Docker, Docker Compose, pytest, Ruff, GitHub Actions

GAN Lab TensorFlow

Real-Time GAN Training Observatory

- Engineered a real-time GAN observatory that streams training telemetry to an interactive browser dashboard at approximately 20 FPS using FastAPI, WebSockets, and a background training engine.
- Implemented a dependency-light NumPy GAN from scratch with manual forward propagation, backpropagation, and Adam optimization, supporting live hyperparameter updates without restarting training.
- Built interactive visualizations for generated distributions, discriminator decision boundaries, learned feature spaces, RBF-MMD, precision, recall, and automatic mode-collapse detection.
- Developed reproducible experiment tracking and model serving with SQLite, allowing users to save configurations, seeds, metric histories, and generator weights, then serve new samples through REST APIs.
- Containerized the platform with Docker and established automated quality gates using GitHub Actions, pytest, and Ruff, while retaining TensorFlow/Keras support for larger MLP and DCGAN experiments.

Tech stacks: Python, NumPy, TensorFlow/Keras, FastAPI, WebSockets, SQLite, JavaScript, Docker, GitHub Actions

RustV OS Lab

Bare-Metal RISC-V Operating System

- Engineered a no_std RISC-V kernel in Rust that boots directly on QEMU without BIOS or SBI firmware, with a custom assembly entry point, stack initialization, boot-hart selection, and machine-mode runtime.
- Implemented trap handling, Sv39 virtual memory, page tables, physical memory protection, system calls, and user-mode fault isolation using full register context frames. • Built a preemptive round-robin scheduler driven by CLINT timer interrupts, with assembly context switching and support for concurrent user-mode tasks.
- Developed a virtio-MMIO block driver from scratch with modern and legacy transport support, split virtqueues, DMA sector operations, feature negotiation, and persistent storage.
- Created TinyFS and an interactive UART shell for persistent file operations, device inspection, and raw-sector access, validated through automated host tests and end-to-end QEMU integration tests.

Tech stacks: Rust, RISC-V Assembly, QEMU, Sv39 Virtual Memory, Virtio-MMIO, DMA, TinyFS, UART, Cargo, GNU Make, GitHub Actions

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Traditional Ethiopian Mancala Game with AI

- Developed a culturally themed Gebeta game for desktop and web using Java Swing and TypeScript, supporting local multiplayer, three AI difficulty levels, responsive controls, animations, and sound effects.
- Built a time-bounded AI opponent using iterative deepening, minimax, alpha-beta pruning, transposition tables, principal-move ordering, and tactical capture prioritization.
- Designed a UI-independent rules engine covering stone sowing, captures, extra turns, illegal moves, scoring, and end-game collection, enabling reliable testing and AI simulations.
- Implemented versioned game persistence for saved sessions and player statistics using Java Preferences and browser localStorage, with defensive corruption recovery.

- Automated unit testing, cross-platform builds, GitHub Pages deployment, and native macOS packaging through Gradle, Vite, GitHub Actions, and jpackage.

Tech stacks: Java, Java Swing, TypeScript, HTML, CSS, Minimax, Alpha-Beta Pruning, Web Workers, Gradle, Vite, JUnit 5, GitHub Actions, GitHub Pages

SKILLS

Programming Languages ● ● ● ● ●

Python, Rust, Java, TypeScript, JavaScript, SQL, RISC-V Assembly, HTML5, CSS3

Backend & Networking ● ● ● ● ●

FastAPI, Node.js, REST APIs, WebSockets, asyncio, TCP Networking, Redis RESP, Firebase Authentication, Firestore

AI, Machine Learning & Algorithms ● ● ● ● ●

TensorFlow/Keras, NumPy, GANs, Minimax, Alpha-Beta Pruning, Iterative Deepening, Model Evaluation, Data Processing Pipelines, LLM Prompt Design

Storage, Observability & Deployment ● ● ● ● ●

SQLite, Persistent Storage, Prometheus Metrics, GitHub Pages, Native Application Packaging, AWS/GCP Deployment Fundamentals

Systems & Database Engineering ● ● ● ● ●

Operating Systems, Distributed Systems, B+ Trees, MVCC, Raft Consensus, Virtual Memory, Preemptive Scheduling, Concurrency, File Systems, Crash Recovery

Frontend & Application Development ● ● ● ● ●

React, Next.js, Java Swing, Tailwind CSS, Responsive Web Design, Web Workers, Interactive Data Visualization

DevOps, Testing & Tooling ● ● ● ● ●

Docker, Docker Compose, Git, GitHub Actions, CI/CD, pytest, JUnit 5, Ruff, Gradle, Vite, QEMU, GNU Make